4. Impacts from this stratospheric warming are likely to reach Earth's surface in the next month and possibly extend through to January.
5. Apart from warming the Antarctic region, the most notable effect will be a shift of the Southern Ocean westerly winds towards the Equator.
6. For regions directly in the path of the strongest westerlies, which includes western Tasmania, New Zealand's South Island, and Patagonia in South America, this generally results in more storminess and rainfall, and colder temperatures.
7. But for subtropical Australia, which largely sits north of the main belt of westerlies, the shift results in reduced rainfall, clearer skies, and warmer temperatures.

58. The northern Australian monsoon season generally lasts from December to March. It is associated with the inflow of moist west to northwesterly winds into the monsoon trough, producing convective cloud and heavy rainfall over northern Australia. These moisture-laden winds originate from the Indian Ocean and southern Asian waters. The north Australian wet season encompasses the monsoon months but can extend several months on either side.

59. **Sea surface temperatures**

Indian Ocean sea surface temperatures impact rainfall and temperature patterns over Australia. Warmer than average sea surface temperatures can provide more moisture for frontal systems and lows crossing Australia.

Indian Ocean Dipole

Sustained changes in the difference between sea surface temperatures of the tropical western and eastern Indian Ocean are known as the Indian Ocean Dipole or IOD. The IOD is one of the key drivers of Australia's climate and can have a significant impact on agriculture. This is because events generally coincide with the winter crop growing season. The IOD has three phases: neutral, positive and negative. Events usually start around May or June, peak between August and October and then rapidly decay when the monsoon arrives in the southern hemisphere around the end of spring.